

Advanced Database Management Systems (CS-320)

Outline and Lecture Plan

Credit Hours: 3 (2-1)

Weekly Schedule: 2 Theory Lectures + 1 Lab Session

Course Objectives

1. Understand advanced concepts in relational and non-relational database systems.
2. Apply normalization, indexing, and optimization techniques to real-world schemas.
3. Design and implement object-relational and distributed databases using Oracle.
4. Develop PL/SQL procedures, triggers, and views for business logic enforcement.
5. Explore emerging trends including NoSQL, Cloud databases, and AI-integrated DBMS.

Course Learning Outcomes (CLOs)

CLO	Learning Outcome	Bloom's Level
CLO-1	Understanding advance data models, technologies and approaches for building database systems	C2 (Understand)
CLO-2	Applying the models and approaches in order to become enabled to select and apply appropriate methods for a particular case	C3 (Apply)
CLO-3	To develop a database solution for a given scenario/ challenging problem in the domain of database systems.	C3 (Apply)

Weekly Breakdown

- **Theory Plan:** 2 lectures per week
- **Lab Plan:** 1 hands-on session per week

Assessment Criteria

Component	Marks	Description
Quiz	10	Short assessments on key concepts
Sessional	10	Class participation, presentations, attendance
Assignment	15	Problem-solving and implementation tasks
Midterm Exam	25	Theory-based exam covering Weeks 1–8
Final Exam	40	Comprehensive exam covering full course
Total	100	

Class Rules & Expectations

- Students can bring laptops for lab sessions.
- Plagiarism in assignments or projects will result in disciplinary action.

- Respectful behavior and academic integrity are mandatory.
- Late submissions will incur grade penalties unless prior approval is granted.

Attendance Policy

- Minimum required attendance: **75%**
- Students falling below this threshold will be barred from final exam.
- Attendance will be recorded per session and monitored weekly.

Instructor Support

- Office Hours: 8:00-16:00 Monday to Friday
- Email: usmanahmed.cs@tuf.edu.pk
- Feedback and consultation available for project guidance and exam prep.

Lecture Plan

Schedule: 2 Lectures per Week

Duration: 16 Weeks

Week 1: Advanced ERD Modeling & Normalization

Lecture 1: EERD Fundamentals & Complex Relationships

- Weak entities, generalization, aggregation
- ERD notation and tools

Lecture 2: Functional Dependencies & Normal Forms

- 1NF to 3NF
- BCNF and multivalued dependencies
- Decomposition strategies
- Lossless join and dependency preservation

Class Assignment: Normalize a hospital ERD to 3NF and draw schema.

Week 2: Object-Relational Features in Oracle

Lecture 1: Object-Oriented Concepts in DBMS

- Encapsulation, inheritance, polymorphism

Lecture 2: Object-Relational vs Relational Models

- Use cases and performance implications

Week 3: File Organization & Indexing

Lecture 1: File Organization Techniques

- Heap, sequential, hashed, clustered

Lecture 2: Indexing Methods and Design

- B-tree, bitmap, composite indexes
- Cost analysis, index selection strategies

Week 4: Transaction Management in Oracle

Lecture 1: ACID Properties & Transaction Lifecycle

- Atomicity, consistency, isolation, durability

Lecture 2: Transaction Control in SQL

- COMMIT, ROLLBACK, SAVEPOINT
- Exception blocks, rollback strategies

Class Assignment: Write a transaction scenario with SAVEPOINT and rollback.

Week 5: Concurrency Control Techniques

Lecture 1: Concurrency Issues

- Lost updates, dirty reads, phantom reads
- Read Committed, Serializable

Lecture 2: Locking Mechanisms

- Pessimistic vs optimistic locking
- Row-level vs table-level locks
- Deadlock detection and resolution

Week 6: Recovery Techniques

Lecture 1: Recovery Concepts

- Types of failures, recovery phases

Lecture 2: Undo vs Redo Logs

- Log-based recovery, checkpoints
- Flashback query, flashback table, flashback drop

Week 7: Query Optimization & Execution Plans

Lecture 1: Query Processing Overview

- Parsing, optimization, execution

Lecture 2: Cost-Based Optimization

- Statistics, selectivity, access paths
- Explain Plan, optimizer hints

Semester Assignment: Analyze execution plan for a multi-join query.

Week 8: PL/SQL Procedures & Functions

Lecture 1: PL/SQL Basics

- Blocks, variables, control structures
- In, Out, In Out parameters
- Return types and usage

Lecture 2: Error Handling & Modular Design

- Exception handling, modular programming

Week 9: PL/SQL Triggers

Lecture 1: Trigger Concepts

- Before, After, Instead Of Triggers

Lecture 2: Row-Level vs Statement-Level Triggers

- Use cases and performance
- Auditing, validation, enforcement

Week 10: Integrity Constraints & Security

Lecture 1: Types of Constraints

- Primary Key, Foreign Key, Unique, Check, Not Null

Lecture 2: User Management

- CREATE USER, GRANT, REVOKE
- Role creation, privilege hierarchy

Class Assignment: Compare flashback recovery with traditional backup restore.

Week 11: Views & Role Management

Lecture 1: Views in SQL

- Simple, complex, read-only views
- View-based DML operations

Lecture 2: Role-Based Access Control

- Role creation, assignment, auditing

Week 12: Physical Database Design & Tuning

Lecture 1: Physical Design Principles

- Table spaces, storage parameters

Lecture 2: Partitioning Strategies

- Range, list, hash partitioning
- Indexing, caching, statistics

Week 13: Distributed Database Concepts

Lecture 1: Distributed DBMS Architecture

- Fragmentation, replication, transparency
- Remote access, cross-node queries

Lecture 2: Challenges in Distributed Systems

- Consistency, latency, fault tolerance

Week 14: MongoDB & NoSQL Introduction

Lecture 1: NoSQL Overview

- Types: document, key-value, column, graph
- Bson, embedded documents, dynamic schema

Lecture 2: CRUD Operations & Querying

- Insert, find, update, delete

Week 15: Emerging Trends in DBMS

Lecture 1: Cloud Databases & Autonomous DB

- Features, architecture, use cases

Lecture 2: AI & Block chain in DBMS

- Predictive indexing, immutable ledgers

Week 16: Final Project Presentations & Review

Lecture 1: Project Guidelines & Evaluation Criteria

- ERD, schema, PL/SQL, performance

Lecture 2: Course Wrap-Up & Exam Preparation

- Key concepts reviews

Reference Books

1. **Database System Concepts** – Silberschatz, Korth, Sudarshan (7th Edition)
2. **Oracle PL/SQL Programming** – Steven Feuerstein
3. **MongoDB: The Definitive Guide** – Kristina Chodorow
4. **Fundamentals of Database Systems** – Elmasri & Navathe

Additional Reading & Resources

- Oracle Documentation: <https://docs.oracle.com>
- MongoDB Manual: <https://www.mongodb.com/docs/manual>
- DBMS Trends: <https://db-engines.com/en/ranking>
- GeeksforGeeks DBMS Section: <https://www.geeksforgeeks.org/dbms/>